Karri Lakshmi Pranavi

ee24btech11210@iith.ac.in | lakshmipranavi291@gmail.com | +91 7337597738 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.97
Narayana Junior College, Mehdipatnam	Class XII (Telangana Board of Intermediate Education)	2024	97.50%
Narayana School Kukatpally, Medchal District	Class X (Board of Secondary Education, Telangana State)	2022	9.80%

PROJECTS

Analog Lock-in Amplifier:

- Designed the schematic and PCB Layout of the amplifier on **Altium Designer** which can extract weak signals with a particular frequency from an input with large environmental noise.
- Implemented **synchronous detection** using **AD633** (Analog Voltage Multiplier) for multiplying the signals and **ADA4610** (Low Noise JFET op-amp) for amplification and filtering (Second-order Sallen-Key Low-Pass Filter).
- Validated the circuit through simulations on **KiCAD** and **LTSpice**, demonstrating effective recovery of the desired signal.
- Achieved **SNR** was **0.01** upto a desired signal frequency of **50kHz**.

Ultra-Low-Power Smart Watch using E-ink Display:

- Designed the **Schematic and PCB** for Smart Watch using **ESP32-S3** which supports deep sleep and RTC memory retention using **KiCAD** along with the use **Electronic Ink Display**.
- **BMA421** (Accelerometer), **ABS07-32.768KHZ-T** (Oscillator), **MCP73831T-2ACI/OT** (battery charging IC) and **MOSFETs** were used to meet low power dissipation and accurate RTC factors.
- Wi-Fi, Bluetooth, Steps Counter and Weather features firmware was also implemented.
- Ongoing work focuses on debugging the current hardware and optimizing the components and PCB Layout for better applications.

Calibration of MoS2 Nanosheet FET:

- Simulated Multilayer MoS2 nanosheet FET (MoS2 as channel and TiN as gate) in a GAA configuration using **2D Schrodinger-Poisson Solver** and **Gmesh** for drawing the MOSFET structure and plotted the **Transfer Characteristics** (logID vs VGS) and **Band Diagrams** using data (parameters) from various research papers and online sources.
- The **interface traps** caused by **inefficient removal of Sacrificial Layers** caused a very high Subthreshold Swing (about 410uA/um) which wasn't possible to simulate in real time using the solver and the mobility of MoS2 was gravely reduced due to surface degradation.

Characterization of CMOS and NMOS Logic Gates:

- Built universal gates **NAND** and **NOR** using **nMOSFETs** and **pMOSFETs** and plotted transfer characteristics for inverter circuit using **Arduino** (generated PWM from Arduino) to capture the data and **Python** for plotting.
- Also measured **Noise Margin** and **Static Power Dissipation** to find better devices among CMOS-based gates and NMOS-based gates.
- CMOS-based gates had better Noise Margin and Low Power Dissipation.

SKILLS

Programming Languages: C | Python | MATLAB | Verilog | LaTeX

Software and Tools: Altium Designer | KiCAD | Xilinx Vivado | LTSpice | Arduino IDE | GitHub | Gmesh | 1D and 2D Schrodinger-Poisson Solver

RELEVANT COURSES

Electronics: Analog Electronics | Digital Systems | Modeling of Nanoscale MOSFETs | Circuit and Network Analysis | Semiconductor Device Fundamentals | PCB Design and Layout for Electronic Systems | Digital Systems Lab | Electronic System Design Project Lab | Electric Circuits Lab | Analog Lab

Mathematics: Probability and Random Processes | Matrix Theory | Intro to Statistics | Vector Calculus

Signal Processing and Control Systems: Signals and Systems | Control Theory for Circuits

POSITIONS OF RESPONSIBILITY

Core Member of Cepheid, Astrophysics Club: Coordinated telescope observation sessions and actively contributed to the development of the club's **Amateur Radio Telescope** project.

Class Representative: Strengthened communication between professors and students, facilitating efficient academic coordination and administrative processes.

Sunshine Buddy: Held catch-up sessions for students, collaborated in organizing wellness programs to foster peer support and participated in wellness workshops.

EXTRACURRICULAR

Astronomy Olympiad: Secured Centre Top 10% in NSEA-2023 conducted by IAPT.

Cultural Event: Secured Bronze Medal in MILAN'25 for Carroms.